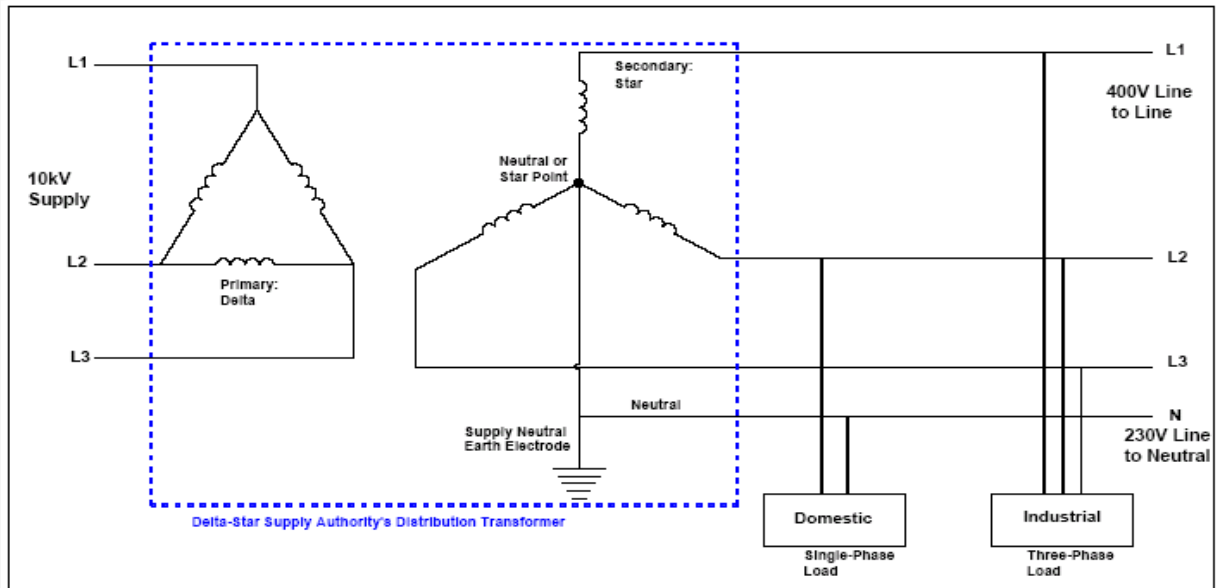
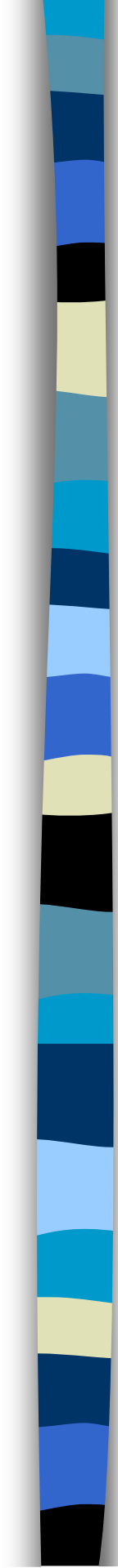


3-phase Supply



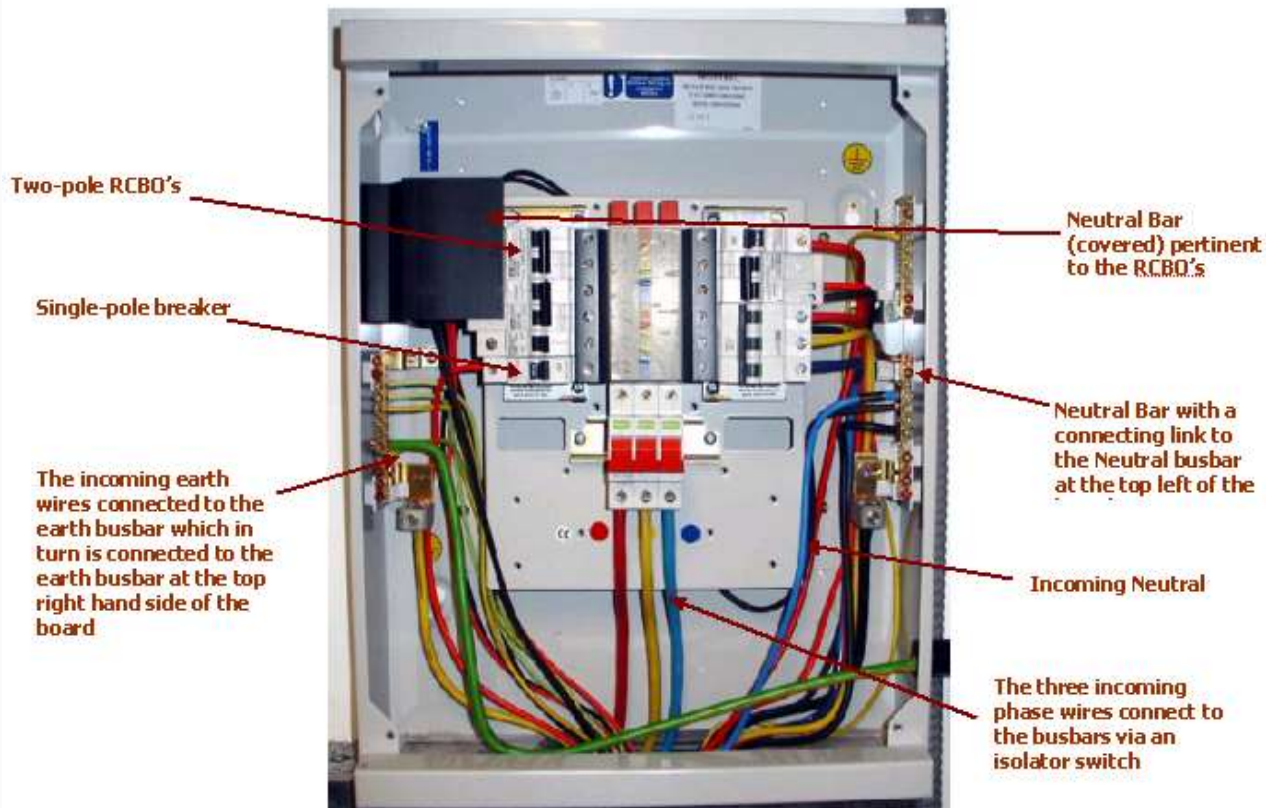
- Division or balancing of Loads
 - Balanced phases
 - Transformer Sizing implications
 - Cable sizing implications
 - Neutral Current implication
- Advantages of a 3-Phase System:
 - Dual Voltage
 - Machine physicality's
 - Rotational Magnetic Fluxes in Machines
 - Transmission implications



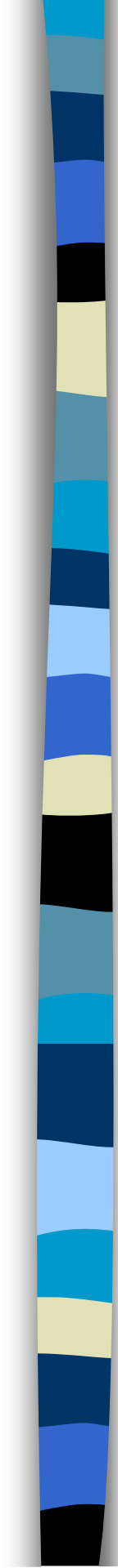
Distribution Boards

- A Distribution Board is described in the ETCI Rules for Electrical Installations (ET101: 2000) as an assembly of protective devices, including two or more fuses or circuit breakers, arranged for the distribution of electrical energy to final circuits or to other distribution boards.
- A distribution board will consist of a suitable enclosure containing suitable facilities for mounting fuses and/or circuit breakers and other protective devices (such as residual current circuit breakers/devices which may, or may not, provide integral overcurrent protection) and other switching and control devices. A distribution board will also contain 'busbars' for interconnecting the circuit breakers or fuses along with neutral and earth bars for connecting the incoming and outgoing neutral conductors and protective conductors. This enclosure may be either of metal clad or all insulated type of construction.

Distribution Boards



The diagram above illustrates a typical 12-position UK distribution panel. It is likely that the manufacturer produces 18 and 24-position versions of this panel using the same chassis which explains why there appears to be so much unused space.



Distribution Boards

■ Protection/ Location of Distribution Boards

- [ET 101: 2000: 538.1, I.S. EN60439]

- Shall be protected against dust, moisture, corrosive or polluting substances, excessive temperatures, impact, vibration and other mechanical stresses.
- Shall be readily accessible and **not** located over cooking or heating appliances, in bathrooms, washrooms or WC's, in storage or airing cupboards, under staircases or where it might be covered by garments.
- Shall **not** be located in an escape route such as a stairway or corridor unless supplementary fire precaution measures are provided. This does not apply to single occupancy buildings.
- Shall not be located above or below, or within 400mm horizontally from a gas meter or a gas appliance in the same space.
- Shall be protected against damage arising from a fault in other services achieved by the use of barriers or by separation.

■ The **Ingress Protection (IP)** rating scheme is an internationally recognised system of denoting the degree of protection afforded by various products against

- Access to hazardous parts and
- Harmful ingress of water.

Distribution Boards

■ Connections

- [ET 101: 2000: 538.1, I.S. EN60439]
- The phase conductors of each two or three phase circuit shall be connected to the same way in a multi-way distribution board.
 - 1st Phase 'Brown' [***must be brown***]
 - 2nd Phase 'Black'
 - 3rd Phase 'Grey'
- Neutral and protective conductors shall be arranged in the same sequence as the corresponding phase conductors.

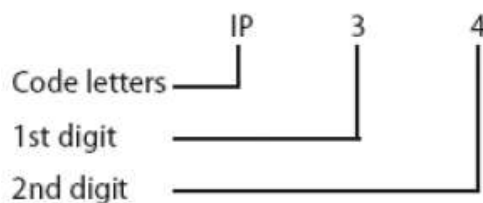
■ Identification & Marking:

- [ET 101: 2000: 514-4 & 515-1-2]
- Protective devices shall be arranged and identified so that the circuits protected may be easily recognised (this being facilitated by labels or other suitable means of identification – no possibility of confusion).
- Record sheets including diagrams and tables shall be available indicating:
 - types of wiring
 - size of conductors
 - rating of protective devices
 - points supplied
 - information identifying protection, isolation and switching devices and their locations.
- Graphical symbols used shall comply with IEC Publication 60617 (Annex 51B ETCI)
- A distribution board not provided with a back plate shall **not** be mounted directly on a combustible surface. A separating material with a flammability rating of FH1 shall be used. These include:
 - plaster board complying with the appropriate standard
 - hardwood such as teak, oak, elm and mahogany.
 - If the mounting surface is of metal it shall be bonded to the protective conductor or to the bonding conductor of the installation.

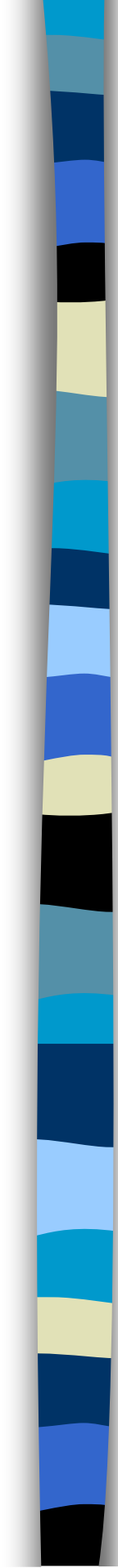
Ingress Protection

IP Level 1st Digit	Description of Protection Level
0	Not protected
1	Protected against solid foreign objects of 50 mm diameter and greater
2	Protected against solid foreign objects of 12,5 mm diameter and greater
3	Protected against solid foreign objects of 2,5 mm diameter and greater
4	Protected against solid foreign objects of 1,0 mm diameter and greater
5	Protected from the amount of dust that would interfere with normal operation
6	Dust tight

IP Level 2nd Digit	Description of Protection Level
0	Not protected
1	Protected against vertically falling water drops
2	Protected against vertically falling water drops when enclosure is tilted up to 15°
3	Protected against water sprayed at an angle up to 60° on either side of the vertical
4	Protected against water splashed against the component from any direction
5	Protected against water projected in jets from any direction
6	Protected against water projected in powerful jets from any direction
7	Protected against temporary immersion in water
8	Protected against continuous immersion in water, or as specified by the user



A component with this IP code indicates that a component is protected against ingress of solid foreign objects having a diameter of 2,5 mm and greater; and is protected against harmful effects due to water splashed against the component from any direction.



Overcurrents – ET101:2000

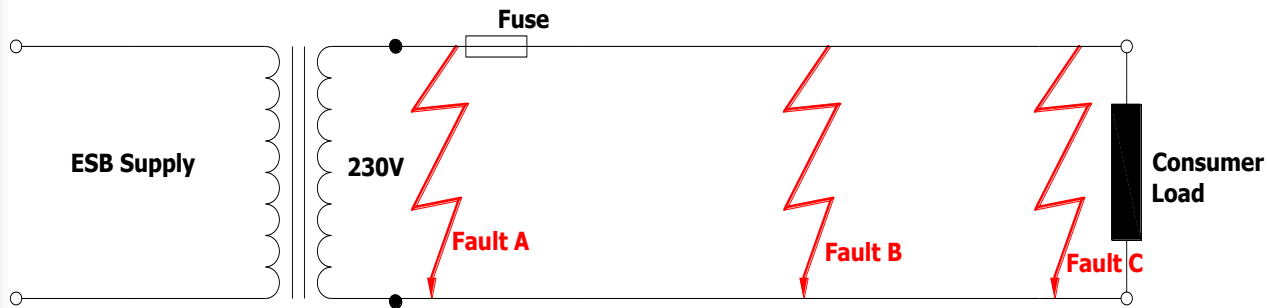
■ Overload:

- An overload current is where too much current is drawn down an electrically healthy circuit e.g. too many appliances are plugged in; there is no fault in the circuit. A properly designed circuit will interrupt an overload before any damage is done to the circuit.

■ Short Circuits

- This is where a fault of negligible impedance (resistance) occurs between live conductors. The value of current, which will flow, will depend on where the fault occurs. Longer runs of cable, particularly smaller cables have a significant attenuating effect on fault current.

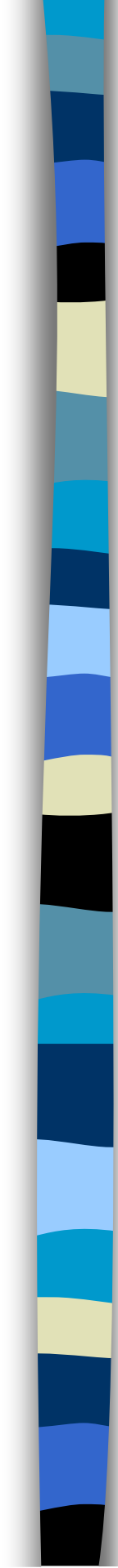
Overcurrents



- The fault level, sometimes known as the prospective short circuit (I_k) is a significant factor when selecting protective devices – particularly circuit breakers.
- The short circuit current at a particular point in an installation is dependent upon:
 - The circuit voltage
 - The total impedance of the circuit including the supply transformer

In large commercial developments, we can reasonably speculate that typical fault levels would be:

- | | |
|------------------------------------|---|
| ➤ At incomer: | 40kA @ p.f. 0.2 |
| ➤ At sub distribution level | 10kA to 3kA
[depending on cable length and cross-sectional area (CSA)] |
| ➤ At distribution level | 1kA to 10kA
[depending on cable length and (CSA)] |
| ➤ At final circuits | 500A to 1kA |



Overcurrents

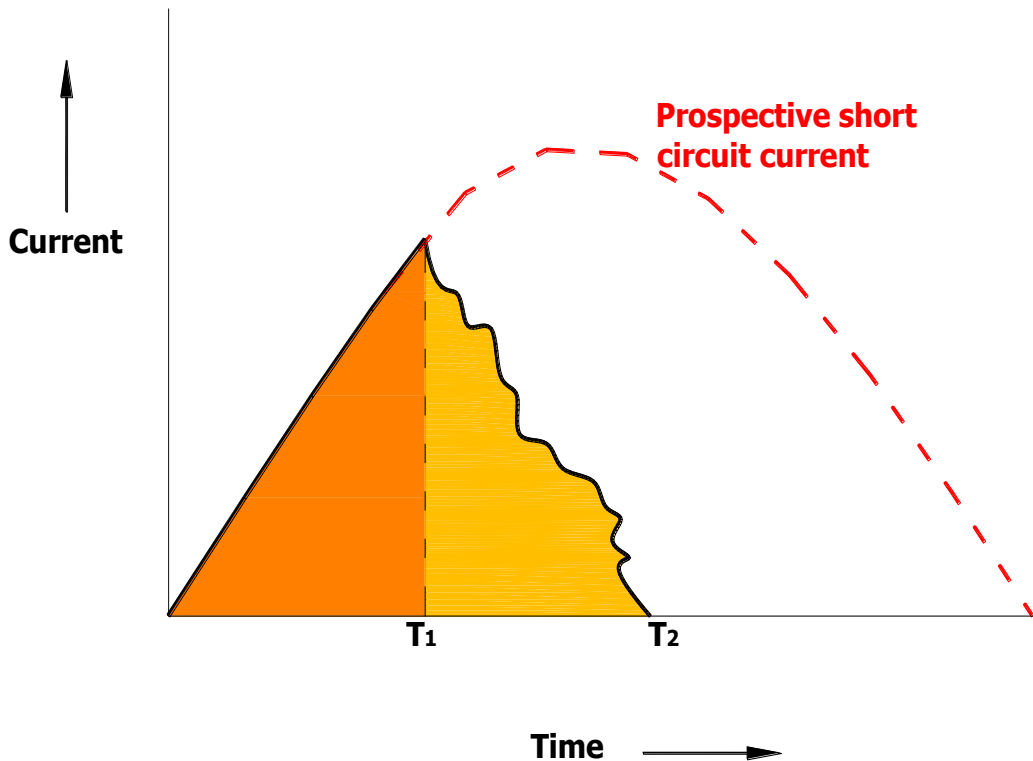
■ **Breaking Capacity :**

- The purpose of determining the short circuit current at a point in an installation is to determine the *Breaking Capacity* in kA of the protective device situated at that point

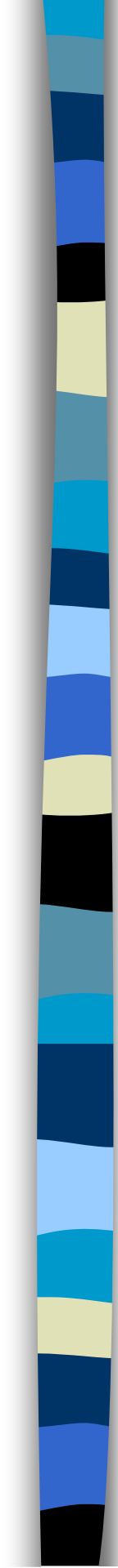
■ **Energy let through in the event of a short circuit is described in terms of:**

- *Pre-arcing Energy*:
 - Energy required to melt the fuse element
- *Arcing Energy*
 - Energy required (post pre-arcing energy) to extinguish the resulting arc

Overcurrents



- The total let through energy is proportional to the energy dissipation during the pre-arcing and arcing intervals and is referred to as the I^2t *characteristic* of the fuse/protective device.

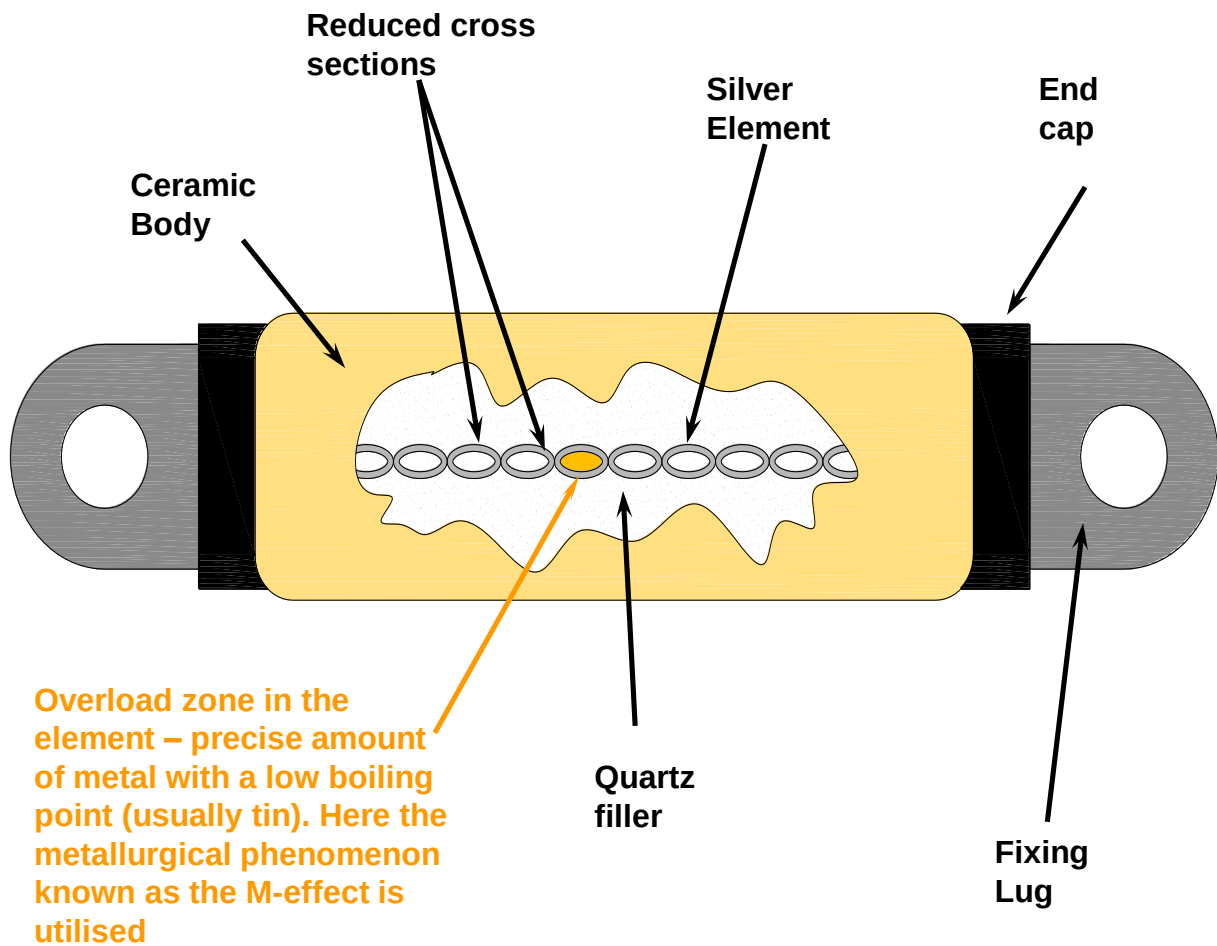


Fuses

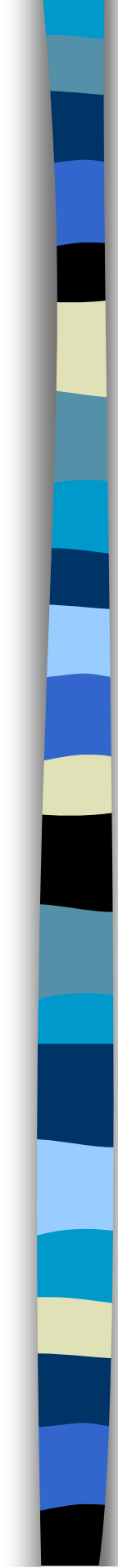
■ Types of Fuses:

- **VDE 0635 DZ type fuse:**
 - This is a cartridge type fuse available in four body sizes D1, D11, D111 and DIV
 - Current ratings from 2 Amps up to 100 Amps.
 - The D1 size is no longer acceptable in this country but may still be found in very old installations.
 - Breaking capacity up to 60kA.
- **VDE 0636 NEOZED or DO type fuse:**
 - This is also a cartridge type fuse available in three body sizes D01, D02 and D03
 - Current ratings from 2 up to 100Amps.
 - Breaking capacity up to 50kA.
- **VDE 0636 NH type fuse:**
 - Breaking capacity of 120kA.
 - They are not designed for replacement by unqualified personnel
 - They are available in ratings up to 1250Amps.
- **BS 1361 fuse:**
 - This is a cartridge fuse available in ratings from 5 to 60 Amps.
 - They are most commonly used in domestic and similar installations and in supply authority cut-outs.
 - They have a breaking capacity of r16.5kA which is adequate for most domestic installation.

High Rupturing Capacity (HRC) Fuses



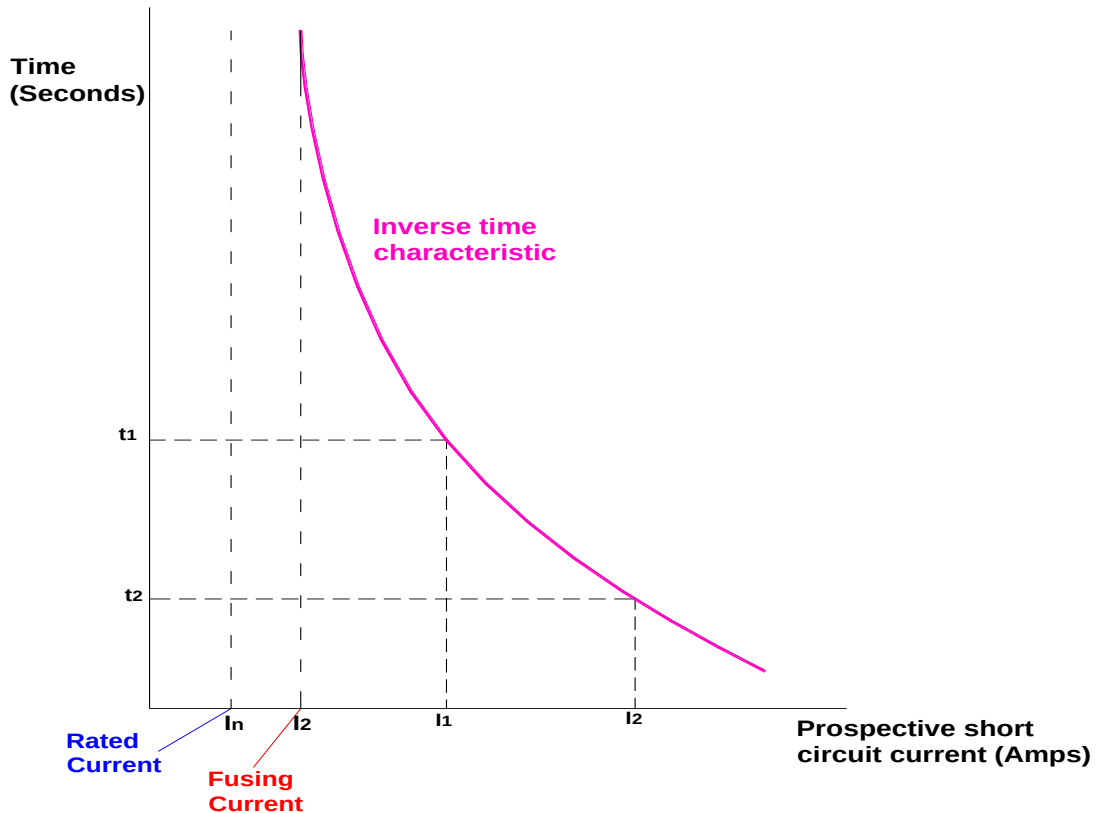
The BS88 HRC fuse consists of a specially shaped silver element totally enclosed in a heat proof body which is filled with very fine grains of quartz. The quartz holds the element in place - even while melting - ensures rapid arc extinction. The element is connected to two tinned brass end caps incorporating fixing lugs as shown above



Advantages of HRC Fuses

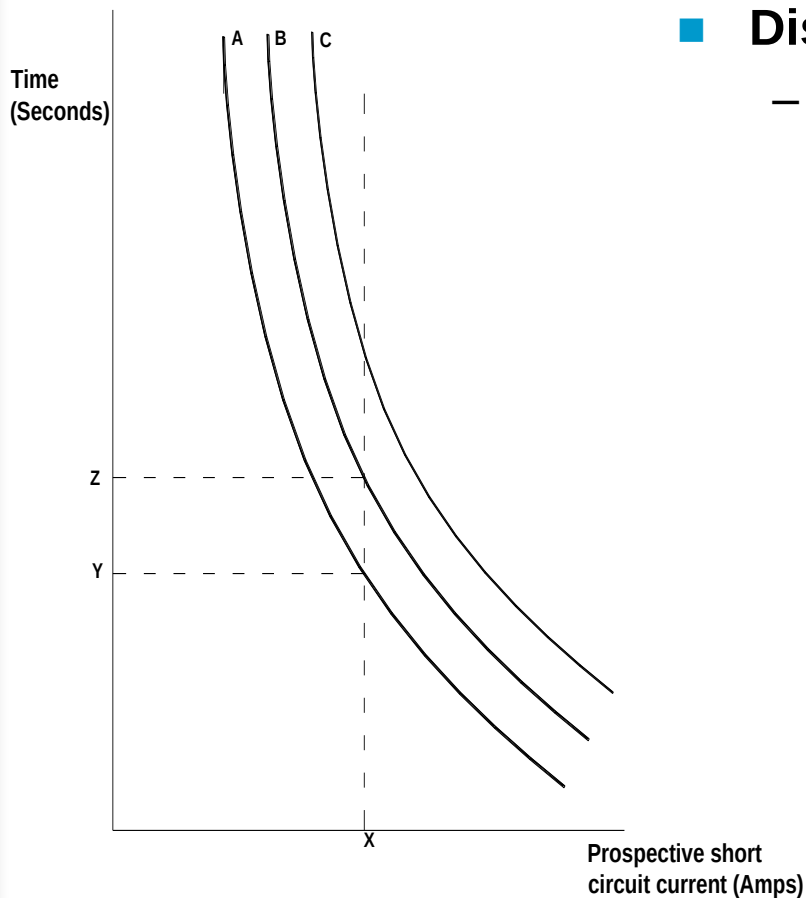
- ✓ Operation is very rapid
- ✓ Capable of breaking very high fault currents safely
- ✓ Declared current rating is very accurate
- ✓ Element does not weaken with age
- ✓ Capable of discriminating between a persistent fault and a transient fault such as the starting of a large inductive motor
- ✓ Different ratings are made to different physical sizes hence they are difficult to interchange

Fuse Characteristics



- For a fuse to satisfactorily protect a cable, its characteristic must match, as closely as possible, the heating characteristic of the cable.
- This means that fuses have an inverse time characteristic, i.e. the larger the over current, the faster the blowing time of the fuse.
- Fuse characteristics are drawn on log/log scale as this enables a wide range of currents along with a wide range of time intervals to be charted

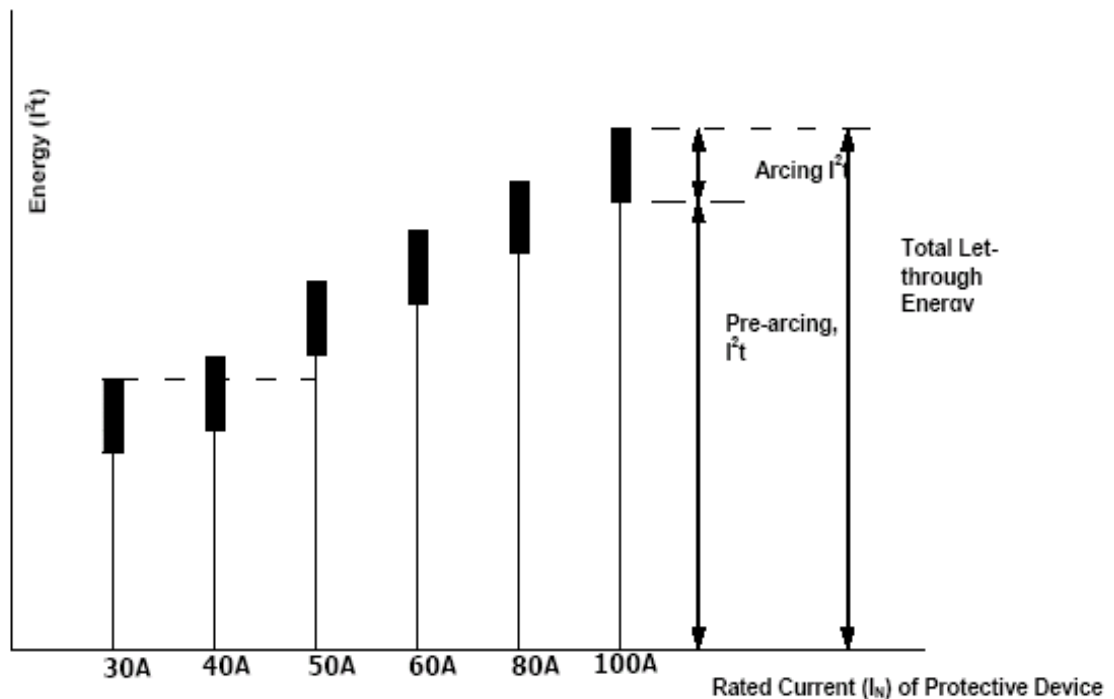
Fuse Characteristics: Discrimination



■ Discrimination:

- In a correctly designed installation, in the event of a fault, the fuse nearest to the fault should interrupt the circuit before any other device has a chance of interrupting it. This is known as discrimination.

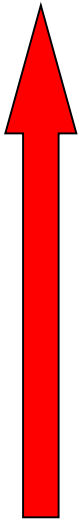
Fuse Characteristics: Discrimination



- As Fuse characteristics will have tolerances associated with their manufacture, it is not possible to rely on Inverse time/current characteristics to design for discrimination.
- It is necessary to use I^2t characteristics

Circuit Breakers

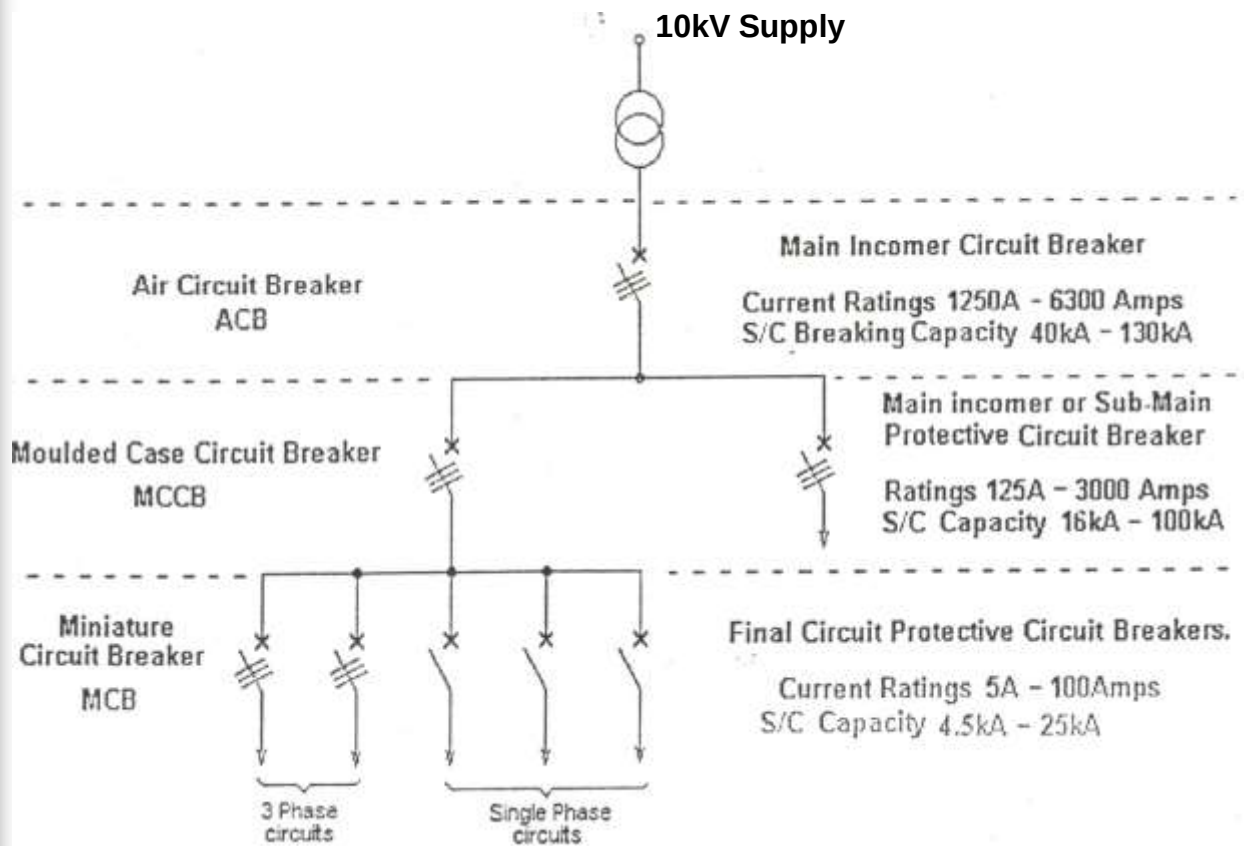
- Circuit breakers are divided into three main types:



- Miniature Circuit Breakers (MCB's)
- Moulded Case Circuit Breakers (MCCB's)
- Air Circuit Breakers (ACB's)

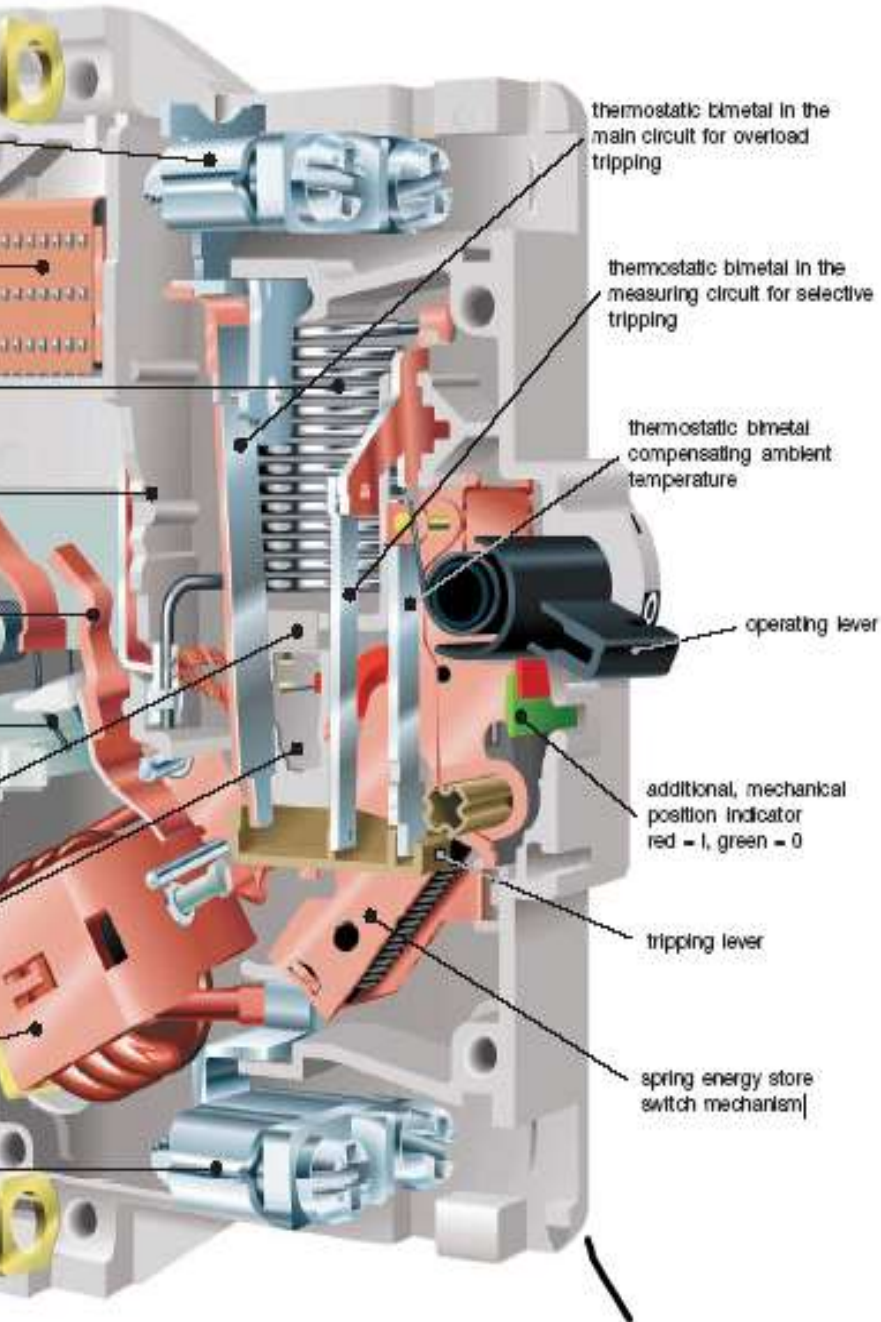
**From Supply
Transformer to
Final Circuits, i.e.
decreasing
breaking capacity**

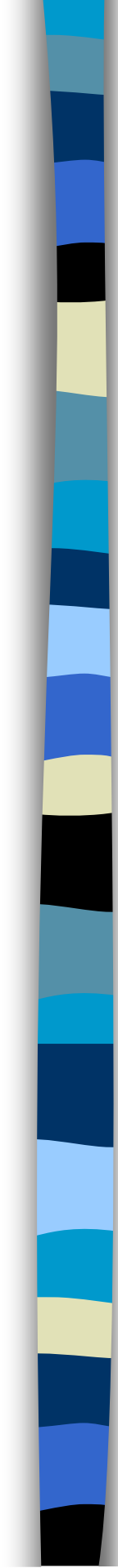
Circuit Breakers



Single line diagram illustrating the sequence in which CBs are employed

Miniature Circuit Breakers





Miniature Circuit Breakers (MCB's)

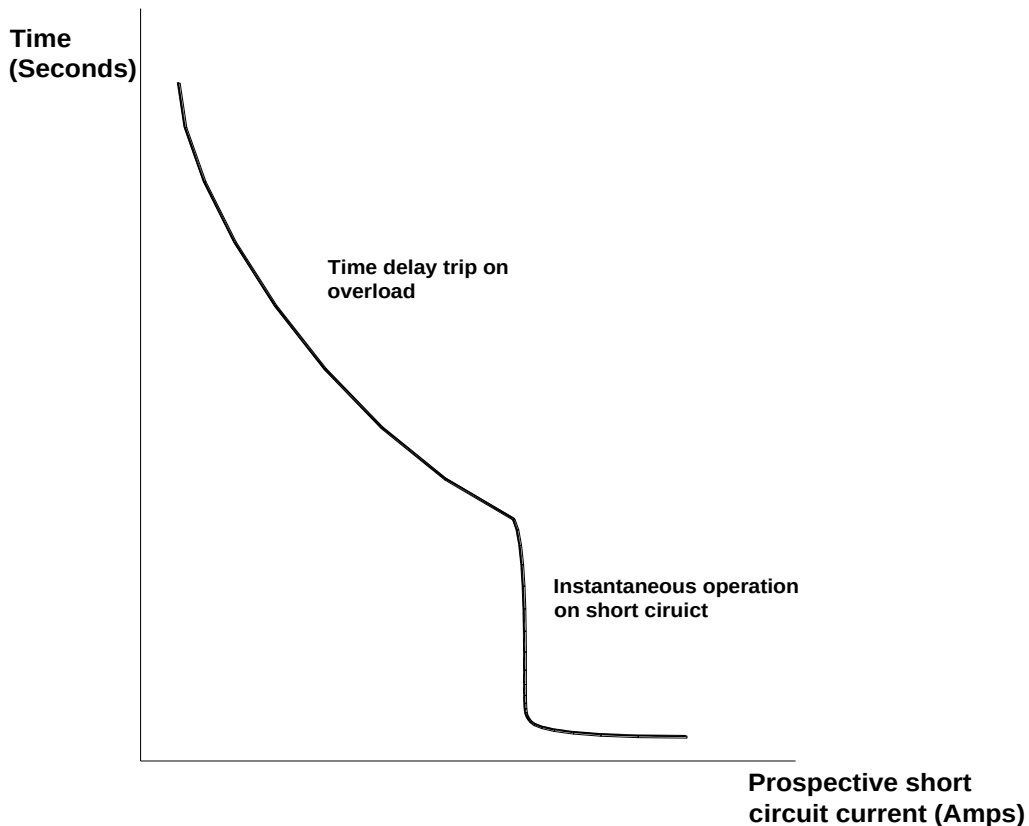
■ Categories of MCB's:

- MCB's manufactured to IS/EN 60898 (IEC 689) are of three types; B,C, D.
- MCB's manufactured to IS/EN 60898 (VDE 0641) are of two types; L and G

■ MCB Overcurrent detection:

- Thermal Tripping
- Magnetic Tripping

MCB Characteristics



■ ***Thermal tripping:***

- In this type of tripping mechanism the current is passed through a bimetal strip connected in series with a magnetic coil.

■ ***Magnetic tripping:***

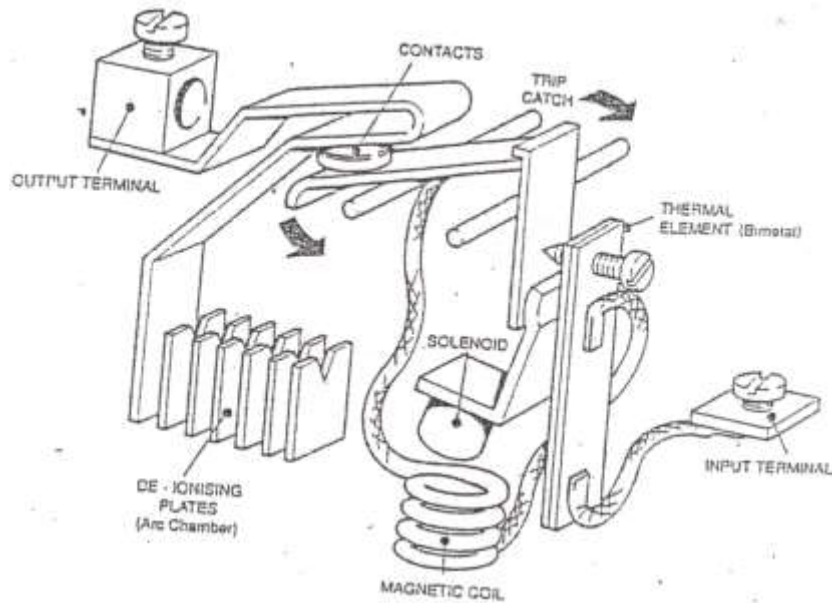
- When a short circuit occurs, the heavy current in the magnetic coil produces a strong magnetic field which instantly opens the breaker

■ ***Arc Extinction:***

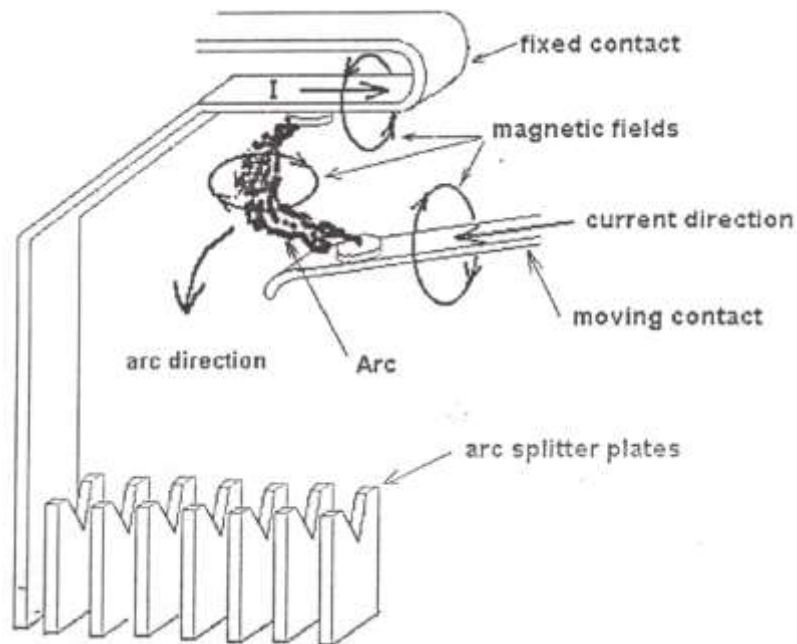
- facilitated by guiding the arc (via self-induced magnetic fields) on splitter plates
- facilitated by guiding the arc (via self-induced magnetic fields) on splitter plates (as illustrated in figure 7). The V-shaped metal splitter plates increase ***the length of the arc, splits it up, cools it and d-ionises it***

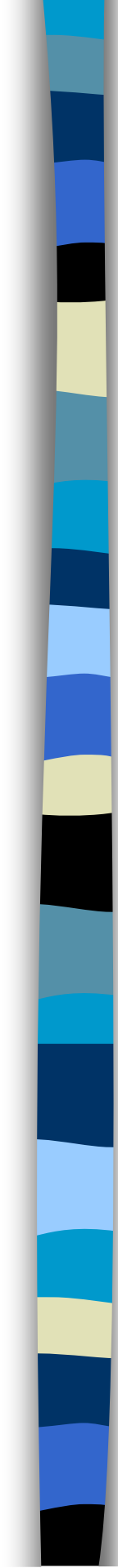
MCB Characteristics

Thermal-Magnetic Tripping



Arc Extinction

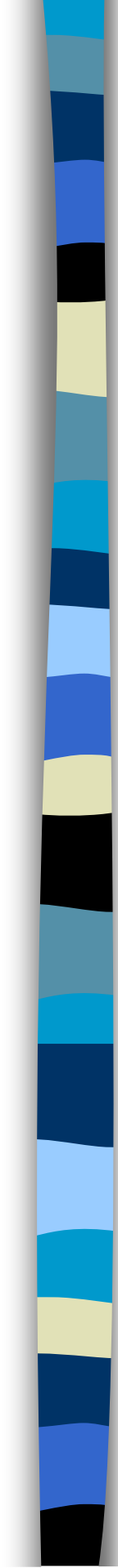




Advantages of MCB's over Fuses

■ ***Advantages of MCB's :***

- ✓ Tripped MCB readily identified even in darkness
- ✓ Cannot be switched back on while fault exists – trip free mechanism
- ✓ Enables supply to be restored immediately and easily even by untrained personnel
- ✓ Accepted as a circuit isolator
- ✓ Locking devices can be attached for maintenance purposes
- ✓ Do not normally require replacement
- ✓ 'Single phasing' of motors is not an issue
- ✓ Do not age in service
- ✓ Tamperproof



Residual Current Devices (RCD's)

- There are two main reasons why RCD's are used:
 - i. To comply with the ETCI rules for electrical installations.
 - i. To provide additional and a higher level of protection than that given by direct earthing, against electric shock and also against fire risk caused by earth leakage currents. Where fuses and miniature circuit breakers (MCB's) are the only means of earth fault protection, it is possible for earth fault currents to flow undetected and cause fire risk (or touch voltage problems).
 - i. The use of an RCD will prevent the flow of a sustained leakage current above the sensitivity of the RCD thus greatly reducing shock and fire risk. RCD's should disconnect all live conductors in the protected circuits in the event of earth leakage current flowing.

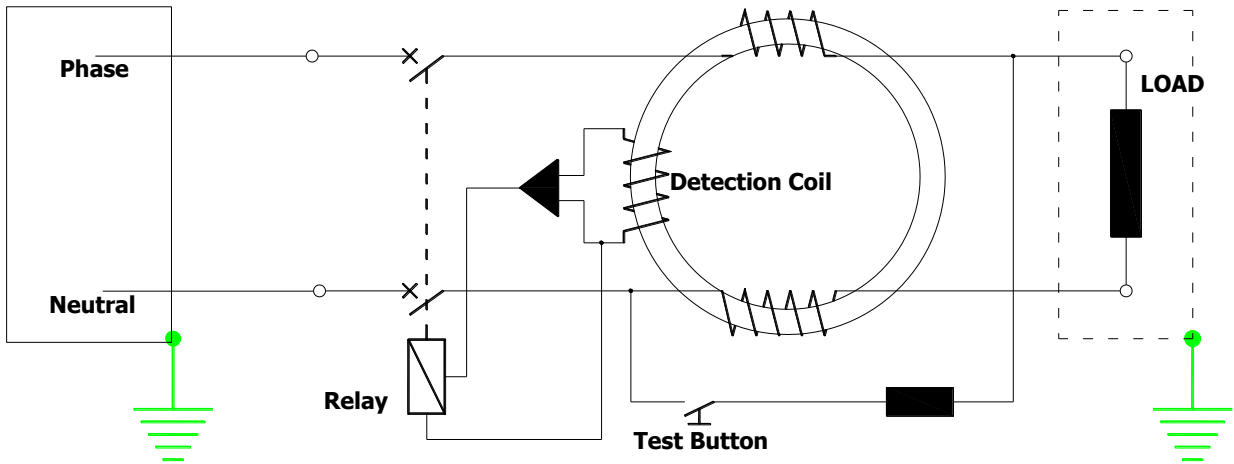
Residual Current Devices (RCD's)

■ Terms associated with RCD's:

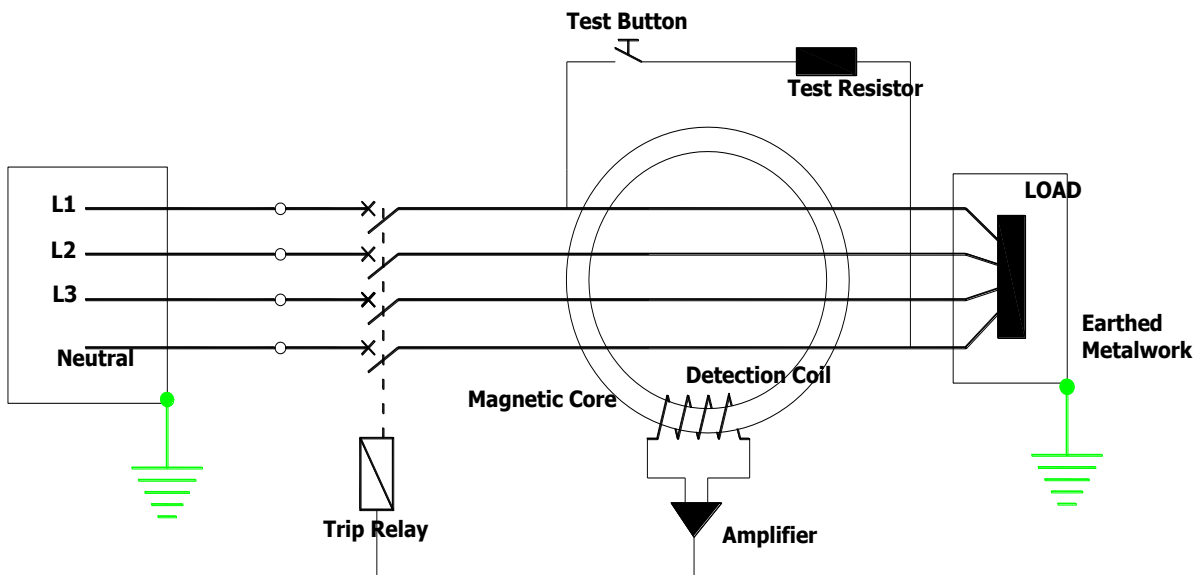
- • **RCCB:**
 - Residual Current Circuit Breaker used in distribution boards to protect individual or groups of circuits
- • **RCBO:**
 - Residual Circuit Breaker with overcurrent protection. This is a
 - combined MCB/RCD and provides overload, short circuit and earth fault protection in one unit
 -
- • **SRCD:**
 - Socket outlet with combined RCD
 -
- • **PRCD:**
 - This is a portable RCD unit with an inbuilt plug top and socket outlet
 -

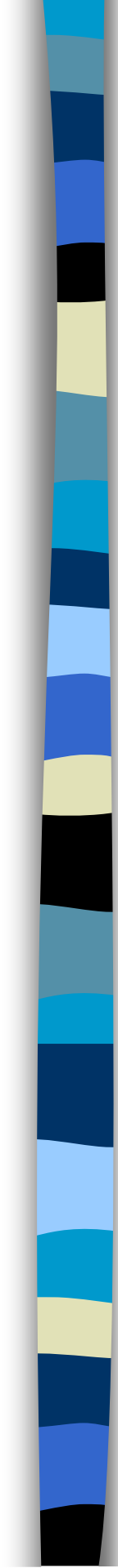
Residual Current Devices (RCD's)

Single Phase RCD



Three Phase RCD

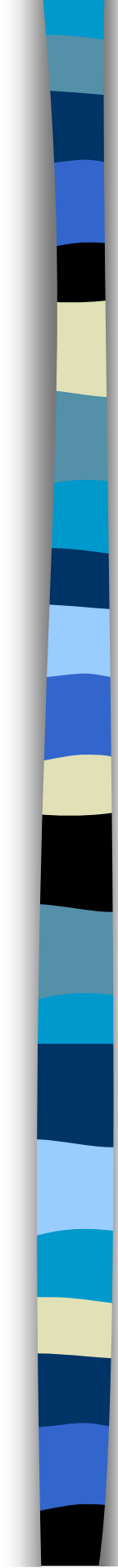




Residual Current Devices (RCD's)

■ Discrimination between RCD's:

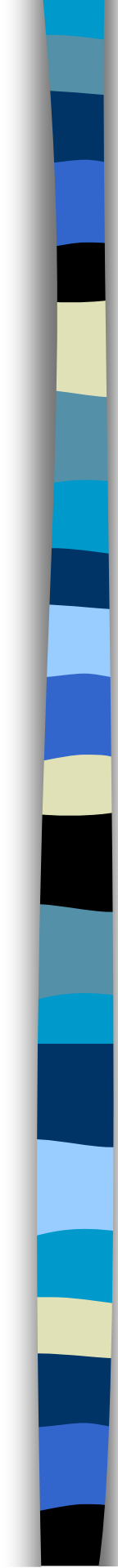
- The time-current characteristic of the device on the supply side shall lie completely above the operating time-current characteristic on the load side
- The rated residual operating current of the device located on the supply side shall be higher than that of the device on the load side
- Selective operation may also be achieved by means of time-delay devices



Residual Current Devices (RCD's)

■ *Nuisance Tripping:*

- Sudden surge of overcurrent
- Voltage spikes/transients
- Inbuilt electronic circuit to protect against such tripping.



Planning Main Switch Boards

- The following information is recommended when determining the size and layout of equipment to be used in a main switchboard:
 - Schedule of all loads (Max demand per phase)
 - Phase balancing of single phase loads
 - Application of diversity
 - Single line block diagram is required
 - Current rating of each item of equipment is included on the block diagram
 - Scaled drawing of the proposed switchboard should be prepared

Planning Main Switch Boards

- Diversity is applied in an installation when determining the values of load current that are likely to be used.
- Diversity is based on assumption that all of the connected load current will not be used simultaneously.
 - E.g. thermostatically controlled devices/equipment and time switch controlled loads are unlikely to demand full loads at all times.
- When determining the current ratings of switchgear diversity can be applied, which will enable a savings to be made in the sizes of cables and in the current ratings of the switchgear. This saves on both cost and spaces
- Diversity is based on the relationship, therefore, between the total load current that is available and the assumed load current demand of an installation.

$$\text{Diversity} = \frac{\text{Assumed Load current}}{\text{Total Load Current}}$$

- Table A31-A Annex 31 A in the ETC/Riles and **Table J1 of the IEE Guidance Notes on the Selection and Erection of Main Switchgear** (more comprehensive guidance)